



BSc (Hons) in **Information Technology**
Specialized in AI

IT3012 : Intelligent Agents

Year 3 · Semester 1 · 2026 Semester 2

Faculty of Computing

Practical 04

Module Objective: In previous weeks, your Drow Ranger agent used A* Search to find a physical path to the goal. However, it blindly assumed every non-wall tile was safe. Today, we implement a **Knowledge Base (KB)** and a **Forward Chaining Inference Engine**. Your agent must now mathematically prove a cell is logically *Feasible* before taking a step, using real-time rules about map vision and enemy threats.

Part 1: The Logic Engine Architecture

Step 1.1: Building the Knowledge Base (KB)

Instead of hardcoding hundreds of nested `if-else` statements in our agent's movement logic, we will build a declarative Knowledge Base. You need to design a Python class that stores **Facts** (current percepts) and **Rules** (Horn Clauses).

1. Create a new file named `logic_engine.py` .
2. Implement a `KnowledgeBase` class based on the following structural requirements:

```

CLASS KnowledgeBase:
    // Attributes
    DECLARE facts AS Set           // To store unique string fact
    DECLARE rules AS List         // To store rules as Tuples: (

    // Methods
    FUNCTION tell_fact(fact_string):
        ADD fact_string TO facts

    FUNCTION tell_rule(premise_list, conclusion_string):
        APPEND Tuple(premise_list, conclusion_string) TO rules
  
```

```
FUNCTION clear_facts():  
    EMPTY the facts Set
```

Part 2: Implementing Forward Chaining

Step 2.1: The Inference Algorithm

The agent needs an Inference Engine to deduce new information from its raw sensors. Translate the following Data-Driven **Forward Chaining** algorithm into a Python method inside your `KnowledgeBase` class.

1. Write the `forward_chain()` method.
2. Ensure you use a `while` loop that continues to iterate over the rules until a full pass results in absolutely no new facts being deduced.

```
FUNCTION forward_chain():  
    DECLARE new_facts_added = TRUE  
  
    WHILE new_facts_added == TRUE:  
        new_facts_added = FALSE  
  
        FOR EACH (premises, conclusion) IN rules:  
            IF conclusion IS NOT IN facts:  
  
                // Modus Ponens Check  
                IF ALL premises ARE IN facts:  
                    ADD conclusion TO facts  
                    new_facts_added = TRUE
```

Part 3: Hooking Logic into the Grid Game

Step 3.1: Defining the Game Constraints

Now we translate the game's safety constraints into our logic engine using Propositional Logic.

1. In your main `agent.py`, instantiate the KB in the `__init__` method: `self.kb = KnowledgeBase()`.
2. Use your `tell_rule` method to define the following safety rules:
Rule 1: `TargetVisible  $\wedge$  HasDust  $\Rightarrow$  SafeToEngage`
Rule 2: `SafeToEngage  $\wedge$  BloodseekerMissing  $\Rightarrow$  Retreat`

Step 3.2: Validating Feasibility in A*

Modify your existing A* pathfinding algorithm to consult the Knowledge Base before expanding a node.

1. Locate the neighbor-generation loop in your A* code (where you normally check for physical walls).
2. Before adding a neighbor to the `open_list`, clear the KB facts.
3. Feed the current percepts for that specific tile into the KB using `tell_fact()`.
4. Run `self.kb.forward_chain()`.
5. If `'Retreat'` is deduced and exists in `self.kb.facts`, mark the tile as **Infeasible**. Skip it (do not add it to the open list), even if it is physically reachable.

Part 4: Self-Evaluation Test Cases

4.1 Automated Validation

Create a file named `test_logic.py`. Copy and run the following Python assert statements to verify your Forward Chaining engine works correctly mathematically before you attempt to integrate it into the visual grid.

```

from logic_engine import KnowledgeBase

def test_forward_chaining():
    kb = KnowledgeBase()

    # Add Domain Rules
    kb.tell_rule(['TargetVisible', 'HasDust'], 'SafeToEngage')
    kb.tell_rule(['SafeToEngage', 'BloodseekerMissing'], 'Retreat')

    # Test Case 1: Safe Engagement
    kb.clear_facts()
    kb.tell_fact('TargetVisible')
    kb.tell_fact('HasDust')
    kb.forward_chain()
    assert 'SafeToEngage' in kb.facts, "Test 1 Failed: Should deduce Sa
    assert 'Retreat' not in kb.facts, "Test 1 Failed: Should NOT deduce

    # Test Case 2: Unsafe Engagement (Bloodseeker Missing)
    kb.clear_facts()
    kb.tell_fact('TargetVisible')
    kb.tell_fact('HasDust')
    kb.tell_fact('BloodseekerMissing')
    kb.forward_chain()
    assert 'Retreat' in kb.facts, "Test 2 Failed: Should deduce Retreat

    print("✅ All Logic Engine Test Cases Passed!")

if __name__ == "__main__":
    test_forward_chaining()

```

Part 5: Theory-to-Implementation Mapping

Based on the code you just wrote and the lecture on Logical Reasoning, complete the following analytical questions to explain *how* your code implements the theory.

1. Reachability vs. Feasibility (Apply)

In Step 3.2, you modified the A* algorithm's neighbor-generation loop. How does your

new code differentiate between checking if a node is "Reachable" (like a wall collision) versus checking if it is "Feasible"? Explain how the Knowledge Base enforces feasibility.

Reachability (Physical): Checks if the agent can physically step onto a cell without hitting a wall or going out of bounds (`(nx, ny) not in walls`).

Feasibility (Logical): Checks if it is safe or logical to step onto that cell based on game rules and threats.

KB Enforcement: The KB takes the tile's percepts, runs `forward_chain()` , and if a rule deduces `'Retreat'` , the A* algorithm skips the cell even if it is physically reachable.

2. Declarative vs. Procedural Paradigms (Analyze)

Why did we build a `KnowledgeBase` class with a `forward_chain()` loop instead of simply writing `if 'TargetVisible' in percepts and 'HasDust' in percepts:` directly inside the agent's movement code? What happens to the agent's code complexity if we add 50 new game rules?

Declarative (Knowledge Base): Keeps game rules separate from movement logic. You only need to define *what* the rules are using `kb.tell_rule()`.

Procedural (if-else): Hardcodes decisions directly inside the movement code using nested conditions.

Code Complexity with 50 Rules: With procedural code, adding 50 rules creates messy, unmaintainable "spaghetti code". With the declarative KB, adding 50 rules requires no changes to the agent's movement code—you just add them to the KB.

3. Modus Ponens & Horn Clauses (Understand)

The rule `['TargetVisible', 'HasDust'] ⇒ 'SafeToEngage'` is formally known as a Horn Clause. Briefly explain how your Python implementation of the `forward_chain()` while loop acts as a mathematical execution of the *Modus Ponens* inference rule.

The Rule: If A is true, then B is true. Since A is true, B must be true.

In Code:

1. `all(premise in self.facts)` checks if all conditions (A) are met.

2. `self.facts.add(conclusion)` automatically adds the result (B).

Finalize and Lock Lab 05 Implementation

✅ **Lab 05 Theory-to-Implementation Mapping completed!**



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